Quiz 11: Blood Spatter & Glass

Review of Session 11. ~15 minutes. Calculator with arcsin needed.

Section A: Spatter shape & velocity

1. A drop hits at an angle and makes an elongated stain with a pointed tail. Which way was the drop traveling?
2. Match the velocity category to the cause:
    - large drops (>4 mm), drips and pools
    - medium drops (1–4 mm)
    - fine mist (<1 mm)
3. What is a "void" in a spatter pattern, and what does it tell you?

Section B: Angle of impact

1. Write the formula for the angle of impact.
2. A stain is 4 mm wide and 8 mm long. What is the angle of impact?
3. A stain is 5 mm wide and 5 mm long. What is the angle of impact?
4. What is the "area of convergence," and how do you find it?
5. What extra information do you need to find the area of *origin*?

Section C: Glass

1. Which cracks form first — radial or concentric?
2. Radial cracks form on which side relative to the force?
3. State the 3R rule.
4. A bullet hole in glass is narrow on one side and wide on the other. Which way was the bullet traveling?
5. Two impact holes: cracks from hole B stop where they meet cracks from hole A. Which impact happened first?
6. A pane with a clean hole and few cracks vs. a shattered pane with many long radial cracks — which was struck by a high-velocity projectile?

Answer Key

Section A

1. In the direction the tail points (and satellite droplets spray forward).
2. Large drops/drips = low (passive) velocity; medium drops = medium velocity (blunt force / stabbing); fine mist = high velocity (gunshot).
3. A blank gap in the spatter where a person or object blocked the blood — it tells you something (or someone) was in that spot.

Section B

1. angle = arcsin(width ÷ length) [sin(angle) = width ÷ length]
2. arcsin(4 ÷ 8) = arcsin(0.5) = 30°.
3. arcsin(5 ÷ 5) = arcsin(1) = 90° (fell straight down / hit perpendicular).
4. The 2-D point the blood came from; draw a line through the long axis of several stains and see where the lines cross.
5. The impact angles of the stains — they give the height (the 3-D origin).

Section C

1. Radial.
2. On the side opposite the force (the reverse/rear side).
3. Radial cracks — the stress ridges meet the glass surface at a Right angle on the Reverse side (the side the force came from).
4. From the narrow side toward the wide side — the cone widens in the direction of travel.
5. Hole A (the crack from B stopped at A's crack, so A's crack was already there).
6. The clean hole with few cracks — high velocity gives the glass no time to flex.